1 **(a)** -8 **[B1]**

(b) Substitutes an x-intercept from the graph, e.g. $4^2 + 4b - 8 = 0$ **[M1]**

-2 **[A1]**

(c) $(1, -9)$ [B2]
(If incorrect, B1 for $x = 1$ or $y = -9$)

[Total Marks: 5]

2 **(a)** Substitutes $(3, 0)$ into $f(x)$ to get $a(3)^3 + b(3)^2 = 0 \implies 27a + 9b = 0$, leading to $3a + b = 0$. **[B1]**

(b) Substitutes $(2, -4)$ to form a second equation: $8a + 4b = -4$ (or $2a + b = -1$) **[M1]**

Solves simultaneous equations (e.g. substitutes $b = -3a$ into $2a + b = -1$) **[M1]**

$a = 1, b = -3$ **[A1]**

(c) Identifies the turning points at $y = 0$ and $y = -4$ as critical values **[M1]**

$k < -4$ or $k > 0$ **[A1]**

[Total Marks: 6]

3 **(a)** 4 **[B1]**

(b) Recognises the period is $120°$ or sets $b \times 120 = 360$ **[M1]**

3 **[A1]**

(c) Recognises there are 2 solutions per period (or sketches horizontal line) **[M1]**

6 **[A1]**

[Total Marks: 5]